

EN 601.124

The Ethics of Artificial Intelligence and Automation:
Frameworks and methods: Fairness

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Announcements

- Reading reflection due tomorrow EOD
 - [Kleinberg, J., Mullainathan, S., & Raghavan, M. \(2016\). Inherent trade-offs in the fair determination of risk scores. arXiv preprint arXiv:1609.05807. \(Up to page 8\)](#)
 - [Mullainathan, S. \(2019\). Biased algorithms are easier to fix than biased people. The New York Times, 6.](#)
- Next week:
 - Quiz on Monday again
 - Fri, Feb 27: Project proposal due
 - Make progress on your project proposal! Use office hours to get feedback!

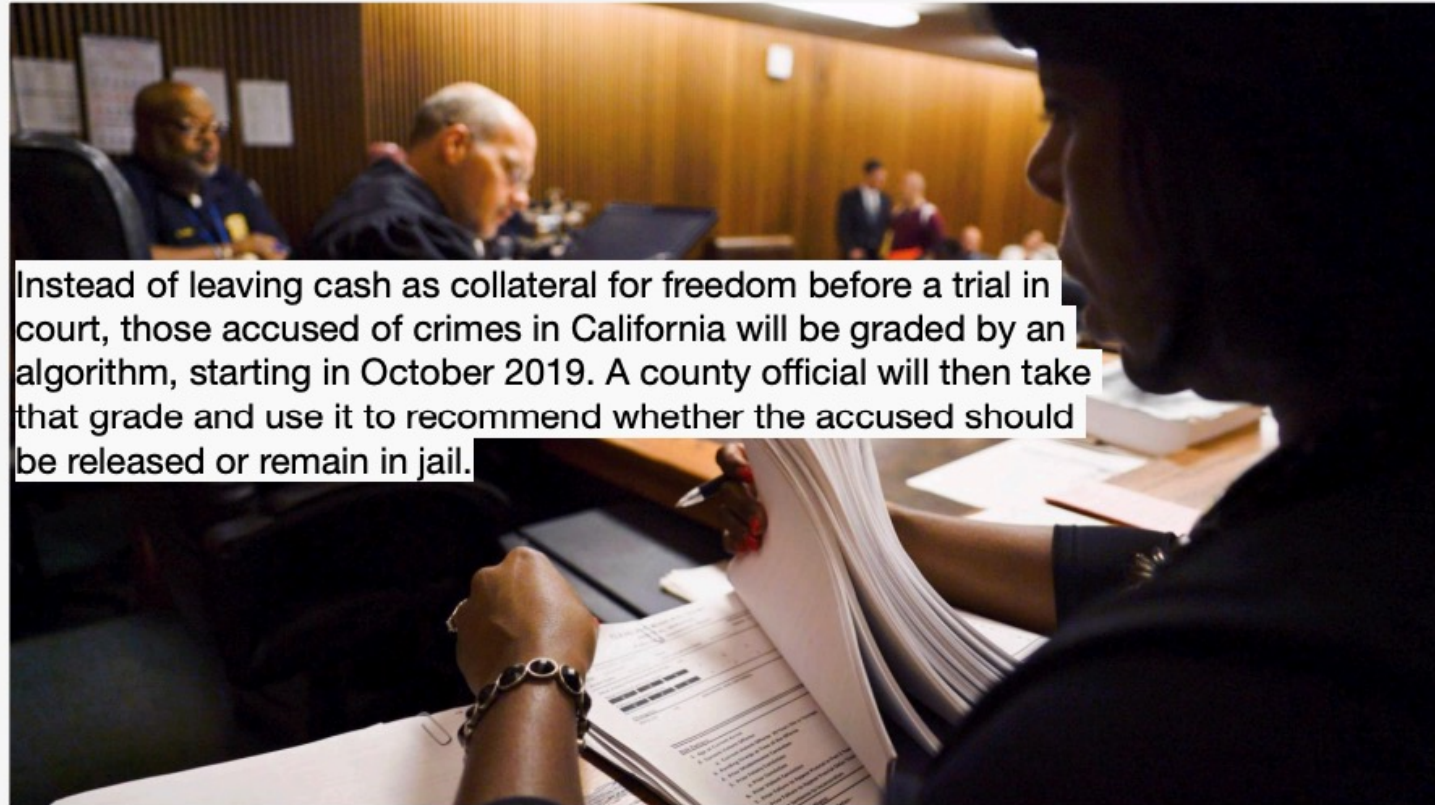
Today's class: Objectives

- Understand how algorithmic decisions shape real-world outcomes
- Recognize that “**fairness**” is not a single concept, but a set of competing definitions
- Identify how **sensitive attributes and proxies** affect algorithmic decisions
- Reason about **trade-offs** between accuracy, equity, and harm

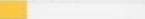
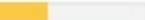
1. **COMPAS analysis**
2. What is fairness in machine learning?
3. Quantitative definitions of fairness in supervised learning
4. Practical tools for analyzing bias
5. Other curveballs

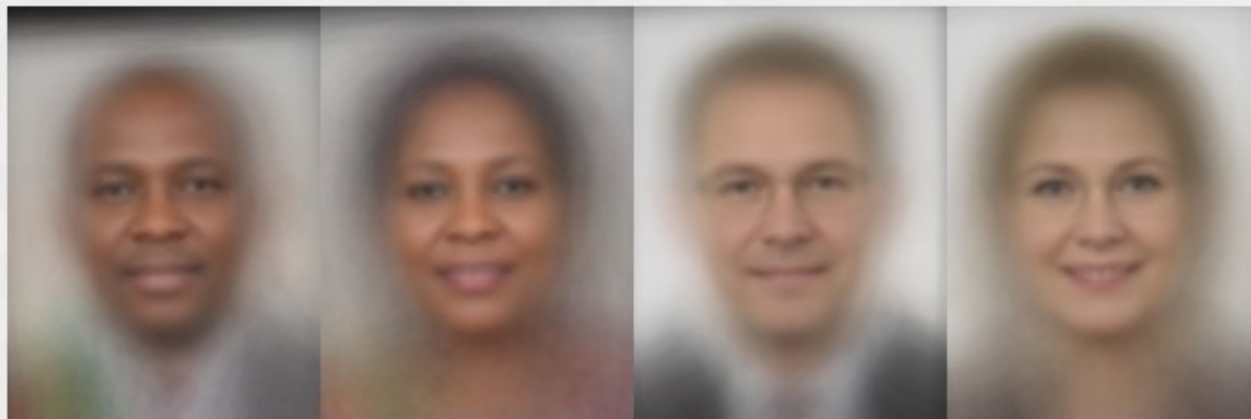
FRYING PAN, FIRE, ETC

California just replaced cash bail with algorithms

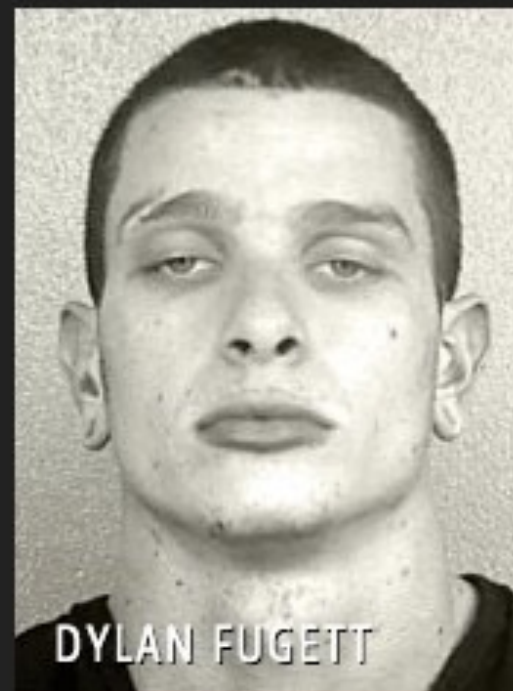
By [Dave Gershgorn](#) • September 4, 2018

A probation officer sorts through automated risk scores in Cleveland.

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
 Microsoft	94.0% 	79.2% 	100% 	98.3% 	20.8% 
 FACE++	99.3% 	65.5% 	99.2% 	94.0% 	33.8% 
 IBM	88.0% 	65.3% 	99.7% 	92.9% 	34.4% 



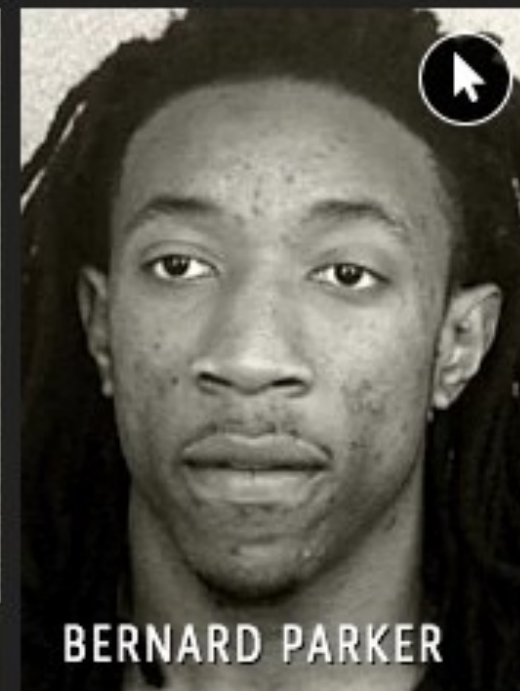
Two Drug Possession Arrests



DYLAN FUGETT

LOW RISK

3



BERNARD PARKER

HIGH RISK

10

Fugett was rated low risk after being arrested with cocaine and marijuana. He was arrested three times on drug charges after that.

<http://gendershades.org/overview.html>

<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>

COMPAS

- ▶ **C**orrectional **O**ffender **M**anagement **P**rofiling for **A**lternative **S**anctions
- ▶ Used in prisons across country: AZ, CO, DL, KY, LA, OK, VA, WA, WI
- ▶ “Evaluation of a defendant’s rehabilitation needs”
- ▶ Recidivism = likelihood of criminal to reoffend

COMPAS

Used on >1M offenders since 2000

- Predicts risk of subject committing a misdemeanor or felony within 2 years from 137 features, not including race
- But, **FP** for blacks was 44.9%, whites 23.5%, **FN** was 47.7% for whites, 28% for blacks in a Broward County study.
- But, COMPAS satisfies
 - **predictive parity** (likelihood of recidivism among high-risk offenders regardless of race)
 - very similar **AUC** for predicting recidivism among whites and blacks
 - provides a **well-calibrated score** regardless of race
- **Human judgment on 7 features achieves nearly identical results**

1. COMPAS analysis
2. **What is fairness in machine learning?**
3. Quantitative definitions of fairness in supervised learning
4. Practical tools for analyzing bias
5. Other curveballs

What is **NOT** bias in machine learning?

- It is not necessarily malicious.
 - Bias can occur even when everyone, from the data collectors to the engineers to the medical professionals, have the best intentions.
- It is not one and done.
 - Just because an algorithm has no bias now does not mean it has no potential later.
- It is not new.
 - Researchers have raised concerns over the last 50 years.

What **IS** bias in machine learning?

- It is defined many ways, for example disparate treatment or impact of algorithm.
- It is the culmination of a flawed system.
 - Sources including bias in the data collection, bias in the algorithmic process, and bias in the deployment.
- It is the vigilance of how technology can amplify or create bias.

Bias, **Technically**

- **{Selection, Sampling, Reporting} bias**
- **Bias of an Estimator**
 - Generally, we have bias, variance, and noise
 - O = optimal possible model over all possible learners (model family)
 - L = best model learnable by this learner
 - A = actual model learned
 - **Bias** = $O - L$ (limitation of learning method or target model)
 - **Variance** = $L - A$ (error due to sampling of training cases)
 - Estimate significance by comparing against learning from randomly permuted data
- **Inductive Bias**
 - assumptions made by the learning algorithm about regularities that allow prediction on unseen cases

Isn't Discrimination the Very Point of Machine Learning?

- Unjustified basis for differentiation
- Practical irrelevance
- Moral irrelevance
- Fairness focuses on ethical concerns
- Discrimination is
 - domain specific — how it influences people's life chances
 - feature specific — socially salient qualities that have served as the basis for unjustified and systematically adverse treatment in the past

What are protected classes?

- ▶ Race (Civil Rights Act of 1964)
- ▶ Sex (Equal Pay Act of 1963; Civil Rights Act of 1964)
- ▶ Religion (Civil Rights Act of 1964)
- ▶ National origin (Civil Rights Act of 1964)
- ▶ Citizenship (Immigration Reform and Control Act)
- ▶ Age (Age Discrimination in Employment Act of 1967)
- ▶ Pregnancy (Pregnancy Discrimination Act)
- ▶ Familial status (Civil Rights Act of 1968)
- ▶ Disability status (Rehabilitation Act of 1973; Americans with Disabilities Act of 1990)
- ▶ Veteran status (Vietnam Era Veterans' Readjustment Assistance Act of 1974; Uniformed Services Employment and Reemployment Rights Act); Genetic information (Genetic Information Nondiscrimination Act)
- ▶ Sexual orientation (Mass SJC 2004, SCOTUS 2015)

Regulated Domains

- ▶ Credit (Equal Credit Opportunity Act)
- ▶ Education (Civil Rights Act of 1964; Education Amendments of 1972)
- ▶ Employment (Civil Rights Act of 1964)
- ▶ Housing (Fair Housing Act)

Two Doctrines of Discrimination Law

Disparate Treatment

- Formal — considering class membership
E.g., country club exclusion based on race or religion,
- Intentional — without explicit reference to class, but with same effect
E.g., red-lining (mortgage availability based on geographic location)

Two Doctrines of Discrimination Law

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Disparate Impact

- Unjustified, Avoidable
- How to demonstrate: “4/5 rule” (20% difference establishes it)
- How to defend: business necessity, job-related

Two Doctrines of Discrimination Law

Disparate Treatment

- Formal — considering class membership
E.g., color, sex, race, national origin, or religion,
- Intentional —
E.g., redlining, but with same effect
on geographic location)



Disparate Impact

- Unjustified, Avoidable
- How to demonstrate: “4/5 rule” (20% difference establishes it)
- How to defend: business necessity, job-related

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How do we define “bias”?

Notation	Meaning
X	Features of an individual (e.g., prior offenses, education, zip code)
A	Sensitive attribute (e.g., race)
Y	Outcome (e.g., grant bail)

How do we define “bias”?

- 1) Fairness through unawareness
- 2) Group fairness
- 3) Calibration
- 4) Error rate balance
- 5) Representational fairness
- 6) Counterfactual fairness
- 7) Individual fairness

How do we define “bias”?

- 1) Fairness through unawareness
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Arvind Narayanan ✓
@random_walker

I wrote up a 2-pager titled "21 fairness definitions and their politics" based on the tweetstorm below and it was accepted at a tutorial for the Conference on Fairness, Accountability, and Transparency!
Here it is (with minor edits):
docs.google.com/document/d/1bn...
See you on Feb 23/24.



Arvind Narayanan ✓ @random_walker · Nov 6, 2017

When I tell my computer science colleagues that there are so many fairness definitions, they are often surprised and/or confused. [Thread]
[twitter.com/random_walker/...](https://twitter.com/random_walker/)

[Show this thread](#)

4:24 PM · Jan 8, 2018 · [Twitter Web Client](#)

60 Retweets 208 Likes

1) Fairness through unawareness

- ▶ Idea: Don't record protected attributes, and don't use them in your algorithm
 - ▶ Predict risk $\mathbf{Y}$ from features $\mathbf{X}$ and group $\mathbf{A}$ using $P(\hat{Y} = Y|X)$ instead of $P(\hat{Y} = Y|X, \mathbf{A})$
- ▶ Pros: Guaranteed to not be making a judgement on protected attribute
- ▶ Cons: Other proxies may still be included in a “race-blind” setting, e.g. zip code or conditions



RESEARCH ARTICLE



Dissecting racial bias in an algorithm used to manage the health of populations

ZIAD OBERMEYER , BRIAN POWERS, CHRISTINE VOGELJ, AND SENDHIL MULLAINATHAN [Authors Info & Affiliations](#)

SCIENCE • 25 Oct 2019 • Vol 366, Issue 6464 • pp. 447-453 • DOI: 10.1126/science.aax2342

172,448 3,914



CHECK ACCESS

Millions of black people affected by racial bias in health-care algorithms

Study reveals rampant racism in decision-making software used by US hospitals – and highlights ways to correct it.

By [Heidi Ledford](#)

- The U.S. health care system uses commercial algorithms to guide health decisions. Obermeyer *et al.* find evidence of racial bias in one widely used algorithm, such that **Black patients assigned the same level of risk by the algorithm are sicker than White patients**
- The authors estimated that this racial bias reduces the number of Black patients identified for extra care **by more than half**.
- Bias occurs because the algorithm **uses health costs as a proxy for health needs**. Less money is spent on Black patients who have the same level of need, and the algorithm thus falsely concludes that Black patients are healthier than equally sick White patients.

Colorblind ideology (CBI)

Colorblind Ideology (CBI)

- The belief that **race should not be acknowledged** in social, political, or institutional decision-making
- Emphasizes “**treating everyone the same**” rather than accounting for structural inequality
- Often framed as **fairness, neutrality, or equality**
- Common phrases: “*I don’t see race*”, “*Race shouldn’t matter*”

Key Assumptions

- Social outcomes reflect **individual effort or merit**, not systemic bias
- Ignoring race will **reduce discrimination**
- Policies that explicitly consider race are **unfair or divisive**

Colorblind ideology (CBI)

Colorblind ideology helps explain why “fairness through unawareness” is appealing but often flawed.

Critiques from Social Science

- **Ignores historical and structural inequality**
- Can **mask ongoing discrimination** by making it harder to name or measure
- Treating unequal groups “the same” can **reproduce inequality**, not fix it
- Shifts focus from systems → individuals

Why This Matters for AI & Tech

- “Race-blind” models may **encode bias implicitly**
- Removing demographic variables does **not guarantee fairness**
- Fairness often requires **awareness, not ignorance**, of social context
- Tension between **formal equality** vs **substantive equity**

2) Group Fairness

- ▶ Idea: Require prediction rate be the same across protected groups
 - ▶ E.g. “20% of the resources should go to the group that has 20% of population”

- ▶ Predict risk Y from features X and group A such that

$$P(\hat{Y} = 1 | \mathbf{A} = 1) = P(\hat{Y} = 1 | \mathbf{A} = 0)$$

- ▶ Pros: Literally treats each race equally
- ▶ Cons:
 - ▶ Too strong: Groups might have different base rates. Then, even a perfect classifier wouldn't qualify as “fair”
 - ▶ Too weak: Doesn't control error rate. Could be perfectly biased (correct for $A=0$ and wrong for $A=1$) and still satisfy.

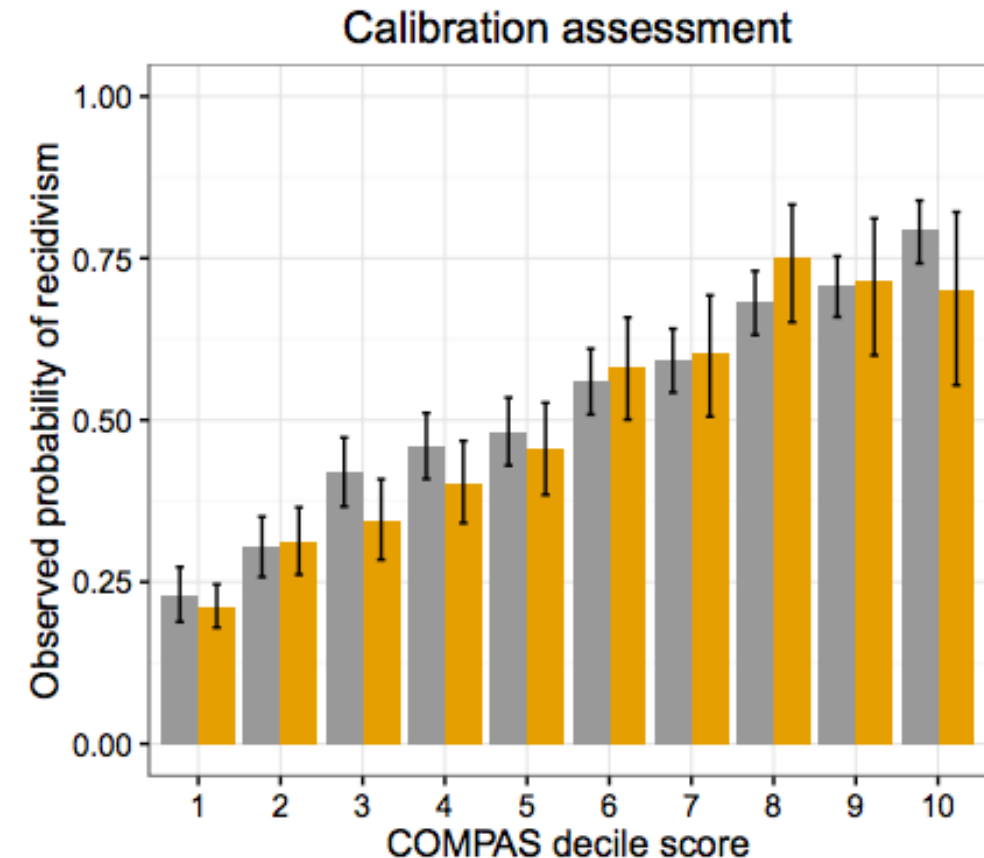
3) Calibration

- ▶ Idea: Same positive predictive value across groups

Predict Y from features X and group A with score S:

$$P(Y = 1|S = s, \mathbf{A} = \mathbf{1}) = P(Y = 1|S = s, \mathbf{A} = \mathbf{0})$$

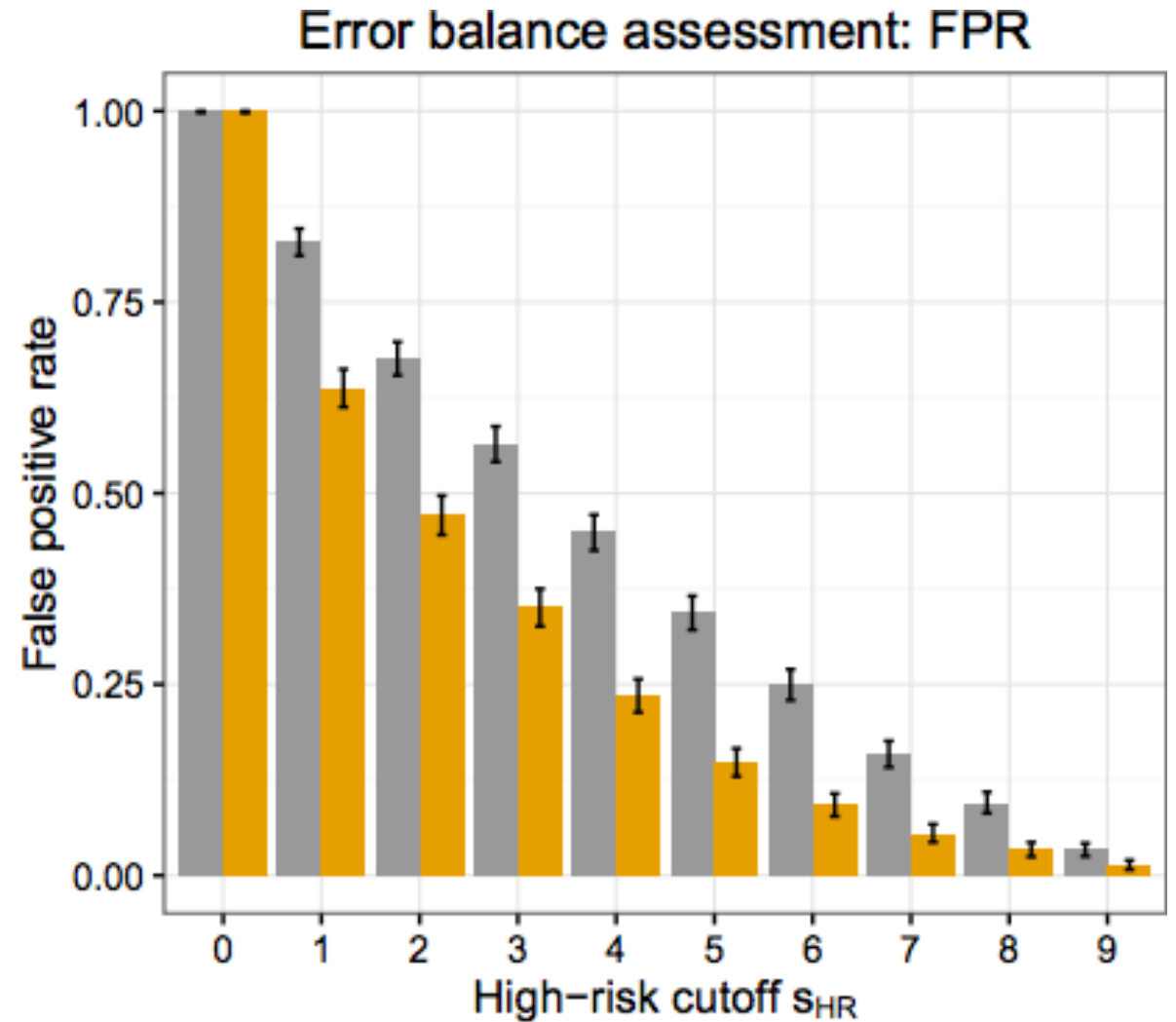
- ▶ Pros: “Equally right across groups”
- ▶ Cons: Not compatible with error rate balance (next slide)



- ▶ Chouldechova, “Fair prediction with disparate impact”, 2017.

4) Error rate balance

- ▶ Idea: Equal false positive rates (FPR) across groups
$$P(\hat{Y} = 1 | Y = 0, \mathbf{A} = 1) = P(\hat{Y} = 1 | Y = 0, \mathbf{A} = 0)$$
- ▶ Pros: “Equally wrong across groups”
- ▶ Cons: Incompatible with calibration and false negative rates (FNR), could dilute with easy cases



Inherent Trade-Offs in the Fair Determination of Risk Scores

Jon Kleinberg *

Sendhil Mullainathan [†]

Manish Raghavan [‡]

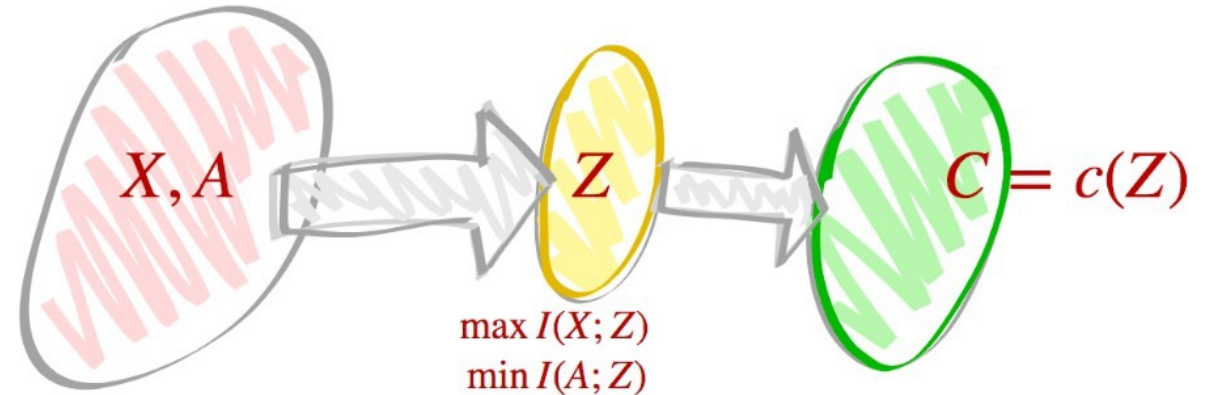
Abstract

Recent discussion in the public sphere about algorithmic classification has involved tension between competing notions of what it means for a probabilistic classification to be fair to different groups. We formalize three fairness conditions that lie at the heart of these debates, and we prove that except in highly constrained special cases, there is no method that can satisfy these three conditions simultaneously. Moreover, even satisfying all three conditions approximately requires that the data lie in an approximate version of one of the constrained special cases identified by our theorem. These results suggest some of the ways in which key notions of fairness are incompatible with each other, and hence provide a framework for thinking about the trade-offs between them.

“We prove that except in highly constrained special cases, there is no method that satisfies these three [fairness] conditions simultaneously.”

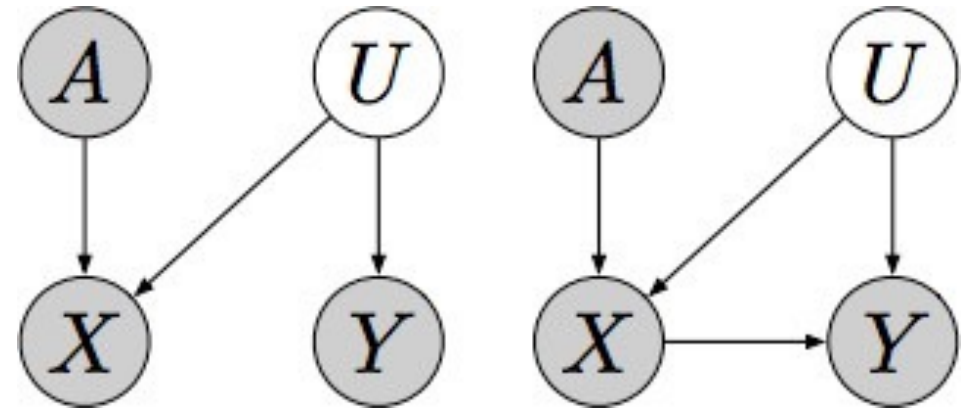
5) Representational Fairness

- ▶ Idea: Learn latent representation Z to minimize group information
- ▶ Pros: Reduce information given to model but still keep important info
- ▶ Cons: Trade-off between accuracy and fairness



6) Counterfactual Fairness

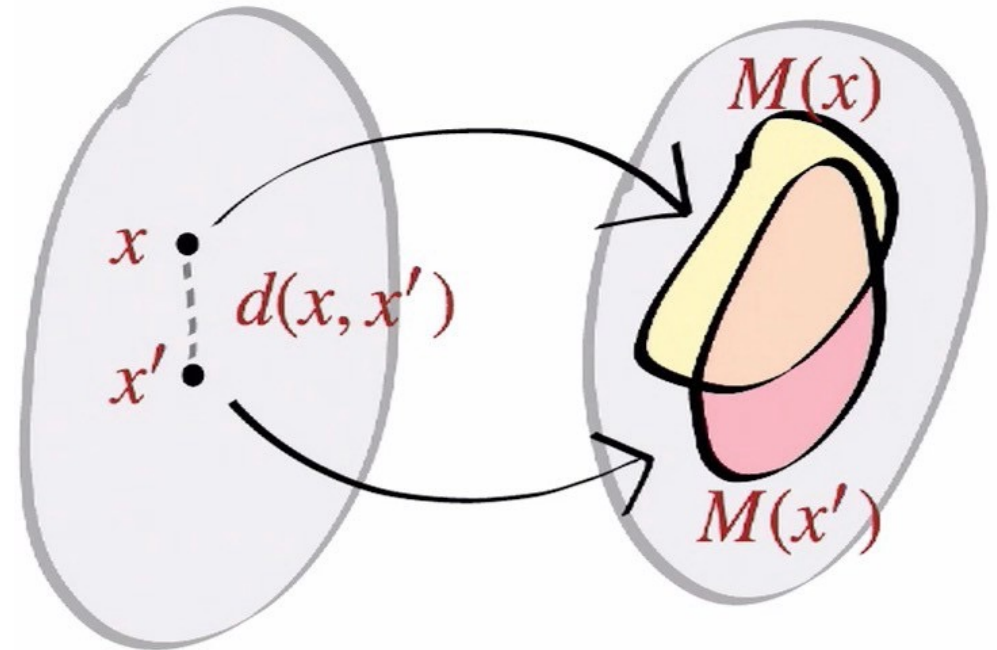
- ▶ Idea: **Group A** should not cause prediction $\hat{Y}$
- ▶ Pros: Can model explicit connections between variables
- ▶ Cons:
 - ▶ Graph model may not actually represent world
 - ▶ Inference assumes observed confounders



$$P(\hat{Y}_{A \leftarrow a}(U) = y \mid X = x, A = a) \\ = P(\hat{Y}_{A \leftarrow a'}(U) = y \mid X = x, A = a)$$

7) Individual fairness

- ▶ Idea: Similar individuals should be treated similarly
- ▶ Pros: Can model heterogeneity within each group
- ▶ Cons: Notion of “similar” is hard to define mathematically, especially in high dimensions



How do we define “bias”?

➤ ~~1) Fairness through unawareness~~

Not useful

➤ 2) Group fairness

➤ 3) Calibration

More standard

➤ 4) Error rate balance

➤ 5) Representational fairness

➤ 6) Counterfactual fairness

More experimental

➤ 7) Individual fairness



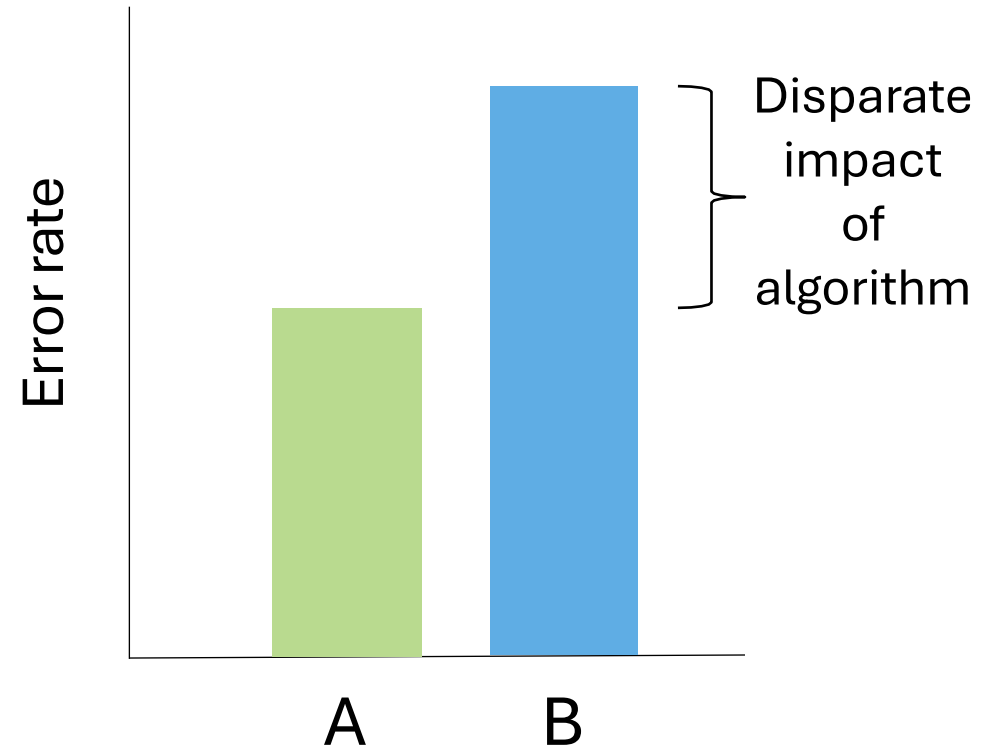
In-class activity

Choose Your Fairness

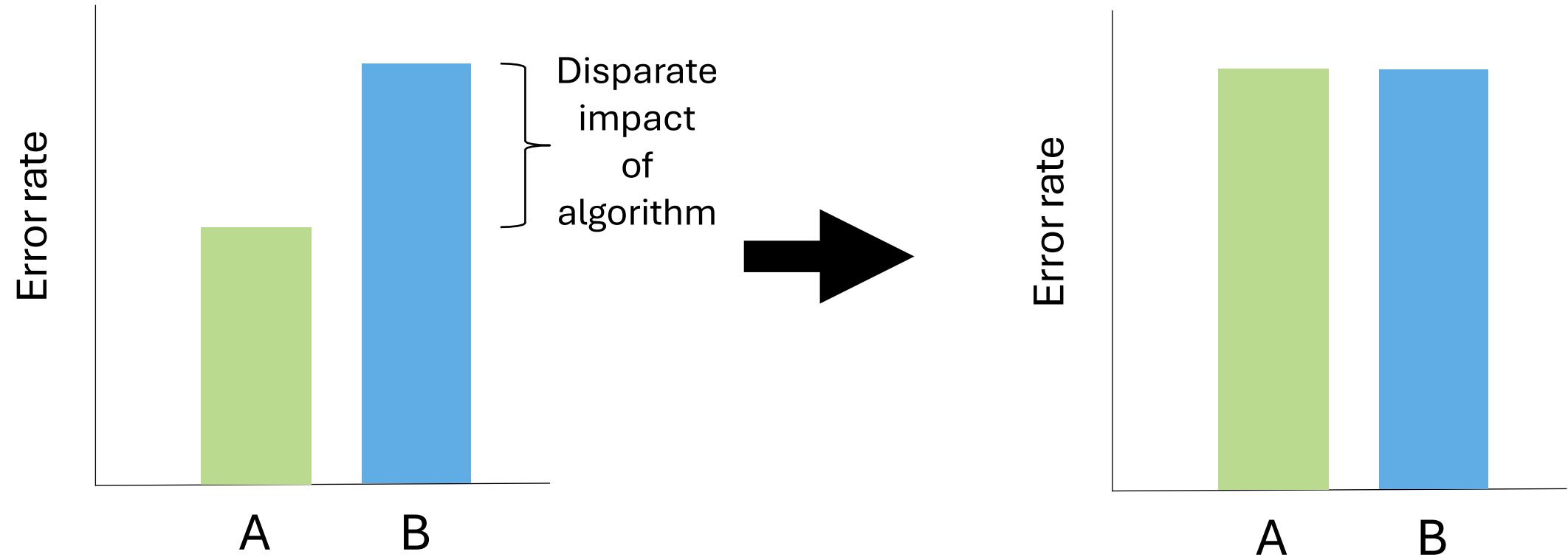
- **Goal:** Internalize trade-offs among fairness definitions.
- **Setup:** Split into groups.
- Each group is assigned one fairness notion:
 - Group fairness
 - Calibration
 - Error-rate balance
 - Representational fairness
 - Counterfactual or individual fairness
- **Task:** Each group answers [10 minutes]:
 - What *value* does this definition prioritize?
 - Who benefits and who might be harmed?
 - In which **domain** (criminal justice, health, hiring, education) is this definition most defensible?

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Tradeoff between accuracy and fairness

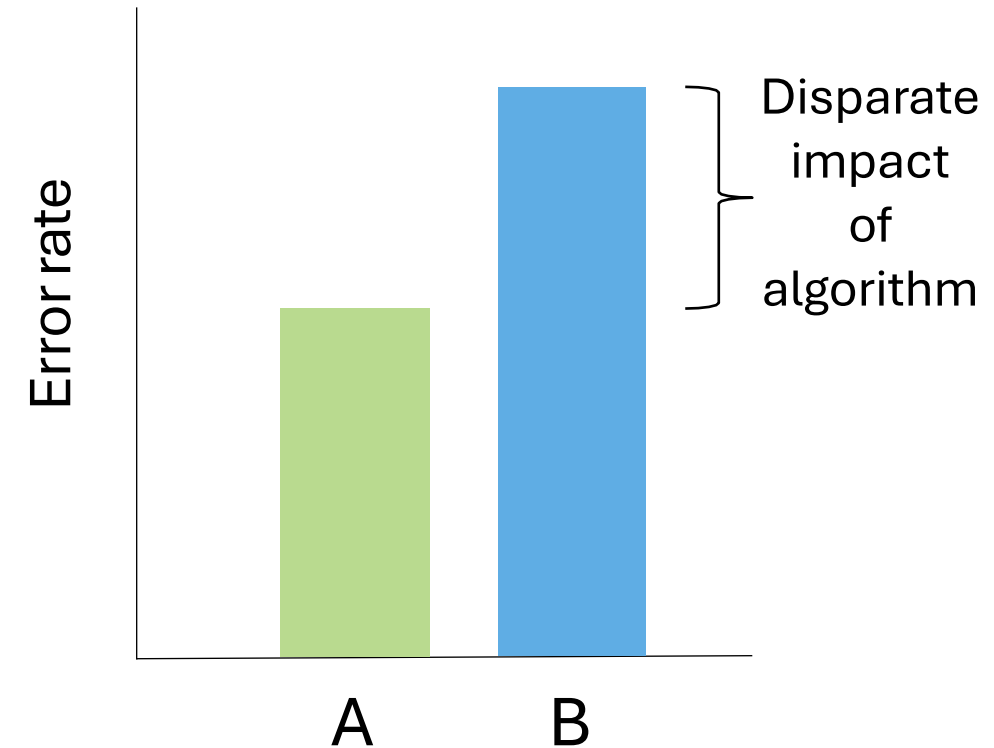


Tradeoff between accuracy and fairness



Consider bias, variance, noise

Bias	How well the model fits the data
Variance	How much the sample size affects the accuracy
Noise	Irreducible error independent of sample size and model



Addressing bias

- Logistic regression: $p(y_i = 1|\mathbf{x}_i, \boldsymbol{\theta}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}_i}}$
- The optimal coefficient: minimize $-\sum_{i=1}^N \log p(y_i|\mathbf{x}_i, \boldsymbol{\theta})$

Fairness Constraints: Mechanisms for Fair Classification

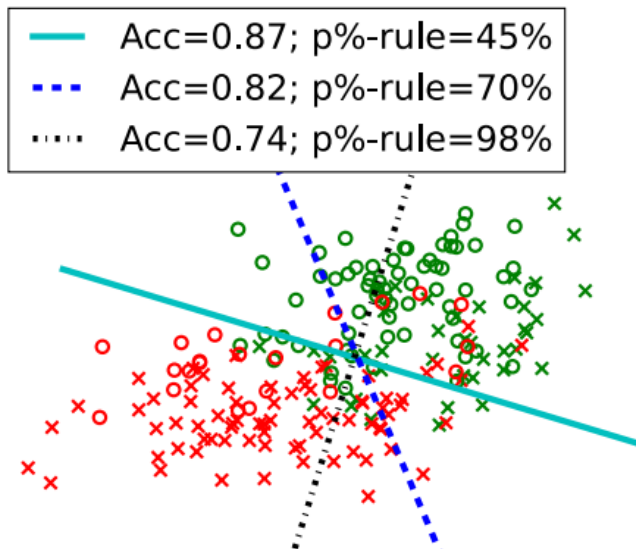
Addressing bias

- Logistic regression: $p(y_i = 1|\mathbf{x}_i, \boldsymbol{\theta}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^T \mathbf{x}_i}}$
- The optimal coefficient: minimize $-\sum_{i=1}^N \log p(y_i|\mathbf{x}_i, \boldsymbol{\theta})$
- With fairness constraints: minimize $-\sum_{i=1}^N \log p(y_i|\mathbf{x}_i, \boldsymbol{\theta})$
subject to $\frac{1}{N} \sum_{i=1}^N (\mathbf{z}_i - \bar{\mathbf{z}}) \boldsymbol{\theta}^T \mathbf{x}_i \leq \mathbf{c},$
 $\frac{1}{N} \sum_{i=1}^N (\mathbf{z}_i - \bar{\mathbf{z}}) \boldsymbol{\theta}^T \mathbf{x}_i \geq -\mathbf{c},$

Fairness Constraints: Mechanisms for Fair Classification

Addressing bias

- Generalizes to SVM & other kinds of classifiers that meet some assumptions



Fairness Constraints: Mechanisms for Fair Classification

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Are Emily and Greg More Employable than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination

Marianne Bertrand, Sendhil Mullainathan

NBER Working Paper No. 9873

Issued in July 2003

NBER Program(s): Labor Studies Program

We perform a field experiment to measure racial discrimination in the labor market. We respond with fictitious resumes to help-wanted ads in Boston and Chicago newspapers. To manipulate perception of race, each resume is assigned either a very African American sounding name or a very White sounding name. The results show significant discrimination against African-American names: White names receive 50 percent more callbacks for interviews. We also find that race affects the benefits of a better resume. For White names, a higher quality resume elicits 30 percent more callbacks whereas for African Americans, it elicits a far smaller increase. Applicants living in better neighborhoods receive more callbacks but, interestingly, this effect does not differ by race. The amount of discrimination is uniform across occupations and industries. Federal contractors and employers who list 'Equal Opportunity Employer' in their ad discriminate as much as other employers. We find little evidence that our results are driven by employers inferring something other than race, such as social class, from the names. These results suggest that racial discrimination is still a prominent feature of the labor market.

The New York Times | <https://nyti.ms/38brSto>

ECONOMIC VIEW

Biased Algorithms Are Easier to Fix Than Biased People

Racial discrimination by algorithms or by people is harmful — but that's where the similarities end.

By Sendhil Mullainathan

Dec. 6, 2019

Open questions

- ▶ How can we build inclusive algorithms and datasets?
- ▶ For what settings should we use algorithms?
- ▶ Can we ever promise an algorithm is “fair”?
- ▶ When should we use humans and when should we use algorithms?

Looking forward

- ▶ AI researchers have made great progress **auditing bias** in existing wide-spread algorithms.
- ▶ **Formalizing fairness** quantitatively can build fairness constraints directly into high-stakes models.
- ▶ Long-term solutions include **growing research community, rethinking datasets, and considering societal impacts.**

Post-class activity: When an algorithm decides for you

- **Goal:** To reflect on how fairness definitions, sensitive attributes, and trade-offs matter when *you* are the subject of algorithmic decision-making
- **Instructions:**
Choose **one real-world scenario** in which an algorithm could meaningfully affect your life. Examples include (but are not limited to):
 - Renting an apartment
 - Loan or credit approval
 - Medical diagnosis or treatment prioritization
 - College admissions or scholarship allocation
 - Job screening or promotion
 - Insurance pricing or coverage decisions
- Assume the algorithm is already widely deployed and cannot simply be “turned off.”
- Answer the **five prompts below** (2–4 sentences each).

Prompt 1: The scenario

- Briefly describe the decision being made and **what outcome is at stake for you**. What would it mean for your life if the algorithm's decision were wrong?

Prompt 2: Sensitive attribute of concern

- Choose **one sensitive attribute** you would personally be most worried about the algorithm using *directly or indirectly* (e.g., race, gender, age, disability, income level, immigration status). Why does this attribute matter in this context? How might it be encoded through proxy variables?

Prompt 3: Fairness metric you want respected

- Choose **one fairness definition or metric** you would want the algorithm to prioritize (e.g., calibration, equal false positive rates, demographic parity, individual fairness). Clearly name the metric.
- Explain **why this notion of fairness feels most important to you in this scenario**.

Prompt 4: Trade-off you are willing to accept

- No system can satisfy all fairness criteria at once. What is **one trade-off** you would accept (e.g., lower overall accuracy, slower decisions, higher costs, unequal error rates)?
- Why is this trade-off acceptable *to you* in this context?

Prompt 5: Trade-off you are not willing to accept

- What is **one trade-off you would reject**, even if it improved overall performance or efficiency?
- Who would be harmed by this trade-off, and why is that harm unacceptable?

References

- Some of the materials adapted from slides by Irene Y. Chen (MIT) and Berkeley's CS 294: Fairness in Machine Learning and N[eur]IPS2017 by Solon Barocas (Cornell) and Moritz Hardt (Berkeley)
- [Angwin, Julia, et al. "Machine bias." 2022. 254-264.](#)
 - [Method](#)
 - [Code](#)

EN 601.124

The Ethics of Artificial Intelligence and Automation:
Frameworks and methods: Fairness

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